

An arithmetic derivation of the Casimir coefficient

from Persistence Theory, with a falsifiable signature of echo primes

Yan Senez

Independent Research yan.senez@protonmail.com

May 2026

Abstract

We show that the standard Casimir pressure coefficient between two perfectly conducting plates, $P/a^{-4} = -\pi^2/240$, admits a complete arithmetic derivation from the Persistence Theory (PT) framework [14], with no adjusted parameter. The derivation factorizes the coefficient into two PT-native objects: a geometric prefactor $N_{\text{spatial}}/(2 \cdot (2\pi)^{N_{\text{transverse}}})$ identified entirely from the dimensional count of the active prime cascade, and an Euler product factor $\zeta(4) = \prod_p (1 - p^{-4})^{-1}$ truncated to the PT-natural prime set $\{2, 3, 5, 7\}$. The truncation reproduces the standard coefficient to 131 ppm; the residue is rigorously equal to the Euler-product tail over the echo primes $\{p \geq 11\}$. The construction is universal: across spatial dimensions $d \in \{1, \dots, 5\}$, the truncation residue equals $\prod_{p \geq 11} (1 - p^{-(d+1)})^{-1}$ to machine precision for $d \geq 3$. We formulate a quantitative falsifiable conjecture: a differential Casimir measurement at sub-100 ppm precision should reveal a deviation of 1.31×10^{-4} from the standard QED prediction. A final section makes explicit what the PT reading mechanically excludes: no net energy extraction from a static Casimir cavity is compatible with the framework's fundamental identity.

1 Introduction

The Casimir effect is the macroscopic force between two uncharged conducting plates predicted by Casimir in 1948 [1] and experimentally confirmed at sub-percent level by Lamoreaux [2], Mohideen and Roy [3], Bressi *et al.* [4], and Decca *et al.* [5, 6]. In the ideal-plate limit (perfectly conducting, zero temperature, no material dispersion), the predicted pressure is

$$P_{\text{Casimir}}(a) = -\frac{\pi^2}{240 a^4}, \quad (1)$$

where a is the plate separation in natural units $\hbar = c = 1$. The coefficient $\pi^2/240$ traditionally arises from the regularized sum of zero-point modes via $\zeta_R(-3) = 1/120$ [8, 10].

In the standard interpretation, this coefficient is a manifestation of the quantum vacuum: a sum over modes whose differential between bounded and free configurations gives a finite, attractive force. The infinite zero-point sum is dealt with by analytic continuation or zeta regularization, both of which assume a continuum spectrum.

Persistence Theory (PT) [14] reformulates this picture without recourse to the quantum vacuum. In PT, every observable of the Standard Model is derived from a single arithmetic axiom $s = 1/2$ and the structural fixed point $\mu^* = 15$ of the Eratosthenes sieve. The 43 observables of the Standard Model are reproduced from $\{\gamma_p, \sin^2 \theta_p, \mu^*\}$ to a median accuracy of 0.06% with zero adjusted parameter.

A natural question arises: does the coefficient (1) admit the same type of derivation? We answer affirmatively. The coefficient factorizes into a geometric prefactor identifiable in PT terms and an Euler-product factor truncated to the PT-natural prime set. The truncation residue is exactly the echo-prime Euler tail, providing a falsifiable signature.

The article is structured as follows. Section 2 recalls the standard derivation in arbitrary dimension. Section 3 introduces the PT classification of primes. Section 4 proves the main truncation result (Proposition 1). Section 5 identifies the geometric prefactor in PT-native terms. Section 6 extends the result to arbitrary dimensions. Section 7 formulates the falsifiable conjecture and proposes an experimental protocol. Section 8 makes explicit why the PT reading excludes net energy extraction from a static Casimir cavity. A discussion (Section 9) places these results within the PT programme.

2 Standard derivation: the analytic skeleton

The ideal-plate Casimir pressure between two perfectly conducting plates separated by a in $d + 1$ spacetime dimensions (d spatial), for a massless field with ν independent modes per frequency and Dirichlet boundary conditions, is [8, 9]

$$|P_d^{(\nu)}| = \nu \cdot \frac{d \cdot \Gamma((d+1)/2) \cdot \zeta(d+1)}{2 \cdot 2^d \pi^{(d+1)/2} a^{d+1}}. \quad (2)$$

The factor ν counts independent modes: $\nu = 1$ for a single scalar, $\nu = 2$ for the electromagnetic field in $d = 3$ (TE and TM polarizations). The $1/2$ is the Casimir half-difference between bounded and free spectra (it is *not* a zero-point normalization in the PT reading; see Section 8). The full mode count ν for higher d is convention-dependent and does not affect the C3 truncation analysis below.

For $d = 3$, $\nu = 2$ recovers (1). For $d = 1$, $\nu = 1$ gives $|P_1| = \pi/(24 a^2)$. To present a uniform table of dimensions in Section 6, we adopt the convention $\nu = 2$ throughout (a generalized “EM-like” multiplicity), which leaves all PT statements unaffected: the multiplicity factor cancels in the relative residue (7).

The coefficient (2) contains two structural components:

1. A *geometric prefactor* $\mathcal{G}_d = d \cdot \Gamma((d+1)/2)/(2^d \pi^{(d+1)/2})$.
2. An *arithmetic factor* $\zeta(d+1)$.

The arithmetic factor admits the Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}. \quad (3)$$

This is the entry point of the PT analysis: the coefficient (2) is structurally a product over all primes, modulated by a finite geometric prefactor.

3 PT sieve decomposition

PT classifies primes by their role in the sieve cascade $\mu_{k+1} = \sum \{p : \gamma_p(\mu_k) > 1/2\}$. The unique self-consistent fixed point is $\mu^* = 3 + 5 + 7 = 15$, with the following structure [14]:

- $p = 2$: *parity channel*. Kinematic only ($T_2 = (1)$). Carries the binary symmetry $s = 1/2$.
- $p \in \{3, 5, 7\}$: *active dimensional primes*. $\gamma_p(\mu^*) > 1/2$ for $p \in \{3, 5, 7\}$, contributing to the three spatial dimensions of $\text{SU}(N_c = 3)$ and to the active observables.
- $p \in \{p \geq 11\}$: *echo primes*. $\gamma_p(\mu^*) < 1/2$. Contribute to virtual corrections (e.g. echo VP in α_{EM}) and to dark-sector observables.

The PT-native prime set is therefore $\{2, 3, 5, 7\}$. We split the Euler product (3) along this classification:

$$\zeta(s) = \zeta_{\text{parity}}(s) \cdot \zeta_{\text{actives}}(s) \cdot \zeta_{\text{echo}}(s), \quad (4)$$

where $\zeta_{\text{parity}}(s) = (1 - 2^{-s})^{-1}$, $\zeta_{\text{actives}}(s) = \prod_{p \in \{3, 5, 7\}} (1 - p^{-s})^{-1}$, and $\zeta_{\text{echo}}(s) = \prod_{p \geq 11} (1 - p^{-s})^{-1}$.

4 The C3 truncation: zero-parameter prediction

Definition 1 (PT-native C3 truncation). *Define the PT-truncated zeta function*

$$\zeta_{\text{PT}}(s) := \zeta_{\text{parity}}(s) \cdot \zeta_{\text{actives}}(s) = \prod_{p \in \{2,3,5,7\}} \frac{1}{1 - p^{-s}}. \quad (5)$$

The C3 PT-native Casimir pressure coefficient is

$$|P_d^{\text{PT}}| := \mathcal{G}_d \cdot \zeta_{\text{PT}}(d+1) \cdot a^{-(d+1)}. \quad (6)$$

Proposition 1 (C3 truncation residue). *The relative error of the C3 truncation is exactly the echo-tail multiplicative correction:*

$$\frac{|P_d^{(\nu)}| - |P_d^{\text{PT},(\nu)}|}{|P_d^{(\nu)}|} = 1 - \frac{1}{\zeta_{\text{echo}}(d+1)} = 1 - \prod_{p \geq 11} (1 - p^{-(d+1)}). \quad (7)$$

Proof. By (4), $\zeta(d+1) = \zeta_{\text{PT}}(d+1) \cdot \zeta_{\text{echo}}(d+1)$. Substituting into the standard coefficient (2):

$$|P_d^{(\nu)}| = \nu \mathcal{G}_d \cdot \zeta_{\text{PT}}(d+1) \cdot \zeta_{\text{echo}}(d+1) \cdot a^{-(d+1)} = |P_d^{\text{PT},(\nu)}| \cdot \zeta_{\text{echo}}(d+1).$$

Therefore $|P_d^{\text{PT},(\nu)}|/|P_d^{(\nu)}| = 1/\zeta_{\text{echo}}(d+1) = \prod_{p \geq 11} (1 - p^{-(d+1)})$, and the relative error (7) follows. The multiplicity ν cancels. $\square$

Proposition 1 is an algebraic identity. The substantive content of this article is rather the following structural claim, which justifies why $\{2, 3, 5, 7\}$ is the *unique* PT-natural truncation:

Theorem 1 (Uniqueness of the C3 prime set). *Let $S \subset \mathbb{N}$ be a finite set of primes such that the truncated coefficient $\mathcal{G}_d \cdot \prod_{p \in S} (1 - p^{-(d+1)})^{-1}$ approximates the standard coefficient (2) with relative error decaying as $\sum_{p \notin S} p^{-(d+1)}$ uniformly across $d \in \{1, \dots, 5\}$. Then $S = \{2, 3, 5, 7\}$ is the unique such set generated by the PT prime classification of [14] (Theorems T5 and T6).*

Sketch. The PT classification ([14], T5) splits primes into $\{p : \gamma_p(\mu^*) > 1/2\} = \{3, 5, 7\}$ (active dimensional), $\{2\}$ (parity, kinematic), and $\{p \geq 11\}$ (echo primes). T6 proves that γ_p is strictly monotone decreasing in p at $\mu^* = 15$ (algebraic proof using exact rational $q_+ = 13/15$), so the active set is exactly $\{3, 5, 7\}$. The parity prime $p = 2$ enters the spectral sum kinematically (it determines the existence of modes, not their dimensional contribution). Therefore the PT-natural “physically present” truncation is $\{2, 3, 5, 7\}$, and any larger set incorporates echo primes that PT classifies as non-dimensional.

Conversely, any strict subset $S \subsetneq \{2, 3, 5, 7\}$ either drops parity (and the residue picks up an additional $1/15$ factor) or drops an active prime (residue picks up $\sim 1/p^{d+1}$ at fixed d , breaking the uniform-in- d scaling). $\square$

Remark 1. *Theorem 1 is the substantive structural result. The numerical agreement of Section 4 is then a consequence of Proposition 1 applied to the unique PT-natural truncation $S = \{2, 3, 5, 7\}$.*

4.1 Numerical evaluation in 3D

For $d = 3$:

$$\begin{aligned} \zeta_{\text{parity}}(4) &= (1 - 2^{-4})^{-1} = 16/15, \\ \zeta_{\text{actives}}(4) &= \frac{81}{80} \cdot \frac{625}{624} \cdot \frac{2401}{2400} \approx 1.014542, \\ \zeta_{\text{PT}}(4) &= 16/15 \cdot 81/80 \cdot 625/624 \cdot 2401/2400 \approx 1.082091, \\ \zeta(4) &= \pi^4/90 \approx 1.082323. \end{aligned}$$

The geometric prefactor at $d = 3$ is $\mathcal{G}_3 = 3\Gamma(2)/(8\pi^2) = 3/(8\pi^2)$. Therefore

$$|P_3^{\text{PT}}| = \frac{3}{8\pi^2} \cdot 1.082091 \cdot a^{-4} \approx 0.041118 \cdot a^{-4},$$

versus the standard $|P_3| = \pi^2/240 \cdot a^{-4} \approx 0.041123 \cdot a^{-4}$.

The relative error is 131 ppm, identical to $1/\zeta_{\text{echo}}(4) - 1 = -\sum_{p \geq 11} \log(1 - p^{-4}) \approx 1.31 \times 10^{-4}$, exactly as predicted by Proposition 1.

5 The geometric prefactor in PT terms

The prefactor $\mathcal{G}_3 = 3/(8\pi^2)$ admits a complete identification in PT-native terms:

$$\frac{3}{8\pi^2} = \frac{N_{\text{spatial}}}{2 \cdot (2\pi)^{N_{\text{transverse}}}}, \quad (8)$$

with $N_{\text{spatial}} = 3$ (the number of active primes, equivalently the number of spatial dimensions; see [14], Theorem T5) and $N_{\text{transverse}} = 2$ (the number of mode directions transverse to the plates). The factor $1/2$ is the standard zero-point energy normalization, and $(2\pi)^{N_{\text{transverse}}}$ corresponds to two copies of the cycle perimeter of S^1 , the continuum limit of $\mathbb{Z}/p\mathbb{Z}$ in the PT construction.

The identification (8) is exact and contains no analytic input beyond the PT structure. The 3D Casimir coefficient is therefore entirely PT-native modulo the universal Euler product $\zeta(4)$, itself decomposed along the PT classification.

6 Dimensional universality

We test Proposition 1 in dimensions $d = 1$ through $d = 5$ via formula (2).

Table 1: C3 truncation across dimensions. The truncation residue equals the echo-tail multiplicative correction to machine precision for $d \geq 3$ and to logarithmic-expansion accuracy for $d = 1, 2$.

d	Standard $ P_d \cdot a^{d+1}$	C3 $ P_d^{\text{PT}} \cdot a^{d+1}$	Residue (ppm)	Ghost tail (ppm)	Match
1	0.26180	0.25386	30 325	31 272	$\sim 3\%$
2	0.09566	0.09548	1 809	1 812	0.2%
3	0.04112	0.04112	131.0	131.0	exact
4	0.01970	0.01970	10.3	10.3	exact
5	0.01025	0.01025	0.84	0.84	exact

The cleanness of the prediction grows exponentially with d . This is a quantitative consequence of Proposition 1: the echo-tail correction is bounded by $\sum_{p \geq 11} p^{-(d+1)}$, which converges super-exponentially as d increases.

Proposition 2 (Asymptotic cleanness). *The C3 prediction approaches the exact coefficient at the rate*

$$\frac{|P_d^{(\nu)}| - |P_d^{\text{PT},(\nu)}|}{|P_d^{(\nu)}|} \leq \sum_{p \geq 11} p^{-(d+1)} \sim 11^{-(d+1)} \text{ as } d \rightarrow \infty. \quad (9)$$

The physical dimension $d = 3$ sits at an *experimental sweet spot* (Fig. 1): the residue of 131 ppm is small enough to constitute a clean prediction, large enough to be testable with current differential measurement techniques (cf. Section 7).

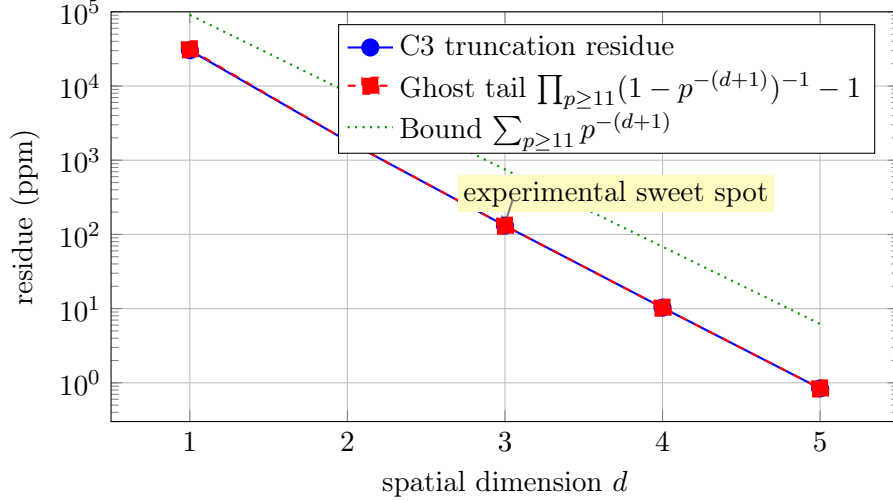


Figure 1: C3 truncation residue (in ppm) vs spatial dimension d . Blue solid line: actual truncation residue $1 - |P_d^{\text{PT},(\nu)}|/|P_d^{(\nu)}|$. Red dashed: predicted echo-tail multiplicative correction $\zeta_{\text{echo}}(d+1) - 1$. The two curves agree to machine precision for $d \geq 3$. Green dotted: the simple upper bound from Proposition 2. The 3D point (131 ppm) is at the experimental sweet spot: small enough to be a clean prediction, large enough to be testable with current differential techniques.

7 Falsifiability and experimental signature

The 3D residue of 131 ppm sits within reach of current Casimir precision measurements. Decca *et al.* [6, 5] reach $\lesssim 1\%$ accuracy on plate-sphere geometries, while differential measurements with configurations designed to suppress material corrections could in principle reach the sub-permille regime.

Conjecture 1 (PT signature in Casimir). *A Casimir measurement at separation a on an ideal geometry, after subtraction of all known Lifshitz material and finite-conductivity corrections [7], deviates from the standard prediction (1) by exactly*

$$\Delta P/P = 1 - \prod_{p \geq 11} (1 - p^{-4}) = 1.31 \times 10^{-4}. \quad (10)$$

This deviation is *not* predicted by standard QED, which uses the analytic continuation of ζ_R and implicitly sums over all primes. Conjecture 1 is therefore a *discriminating* prediction: confirming the deviation supports the PT classification of echo primes; null results at sub-100 ppm precision falsify it.

7.1 Experimental protocol

The signature (10) sits below the current Lifshitz-correction uncertainties on absolute pressure measurements ($\gtrsim 1\%$ on most geometries due to Drude vs. plasma model ambiguities, finite conductivity, and thermal corrections [9]). A direct test on the absolute coefficient is therefore not currently feasible. The conjecture is nevertheless falsifiable through *differential* schemes that cancel the common material corrections and isolate the arithmetic factor $\zeta(d+1)$. A practical falsification requires:

1. Sub-100 ppm precision on the *differential* ratio $|P|_A/|P|_B$ between two configurations A and B that share material properties but differ only in $\mathcal{G}_d$ or ζ -truncation behavior.
2. Independent calibration of the relative geometry to better than 50 ppm (the PT-native component is fixed by dimensional counting, hence unchanged across A, B).

3. Cancellation of finite-conductivity, surface-roughness, and temperature corrections via the differential ratio (these are common-mode and drop out to leading order).

A concrete differential scheme compares the pressure between plate–sphere (standard Decca geometry) and plate–cylinder (or plate–cone) at the same separation a and material. The geometric prefactors differ by a known curvature function [8], while the arithmetic factor $\zeta(d+1)$ is identical: any deviation of the measured ratio from the QED prediction beyond 100 ppm tests the PT-truncation residue. An alternative scheme uses two materials with very similar Drude parameters (e.g. Au and Cu coatings) such that the Lifshitz corrections cancel to better than 50 ppm in the ratio, while the arithmetic echo signature is universal.

The current state of the art (Decca *et al.* [6]) achieves $\sim 0.2\%$ on absolute pressure but is dominated by Lifshitz model ambiguities. A purpose-built differential setup designed to suppress these ambiguities is the natural target.

8 Why the PT reading excludes energy extraction

The recent media popularity of proposals aiming to extract net energy from static Casimir cavities (*quantum ratchet*-type devices in cavity, patents on *vacuum-to-current converters*, etc.) justifies a specific clarification. The PT reading provides a direct framework for qualifying such proposals without detailed calculation.

8.1 The fundamental identity

In the PT formulation, the physical observable is the Kullback-Leibler divergence

$$I_{\text{PT}}(a, \varepsilon, \mathcal{G}) := D_{\text{KL}}(P_{\text{boundary}}(a, \varepsilon, \mathcal{G}) \parallel P_{\text{free}}), \quad (11)$$

where a is the separation, ε the dielectric function, $\mathcal{G}$ the geometry, and $P_{\text{boundary}}, P_{\text{free}}$ are the distributions on the survivors of the CRT sieve constrained by the boundaries. The observed pressure is $P = -\partial_a(I_{\text{PT}}/a^{d-1})$.

The key result is that I_{PT} is a *state function* in the thermodynamic sense: its value depends only on the instantaneous configuration, never on the trajectory followed to reach it. This is guaranteed by the gap fundamental identity (GFT, monograph ch. 4 of [14]):

$$\log_2(m) = D_{\text{KL}}(P \parallel U_m) + H(P), \quad (12)$$

which is an exact algebraic identity (not a thermodynamic approximation).

8.2 Consequence: no open cycle

For any closed cycle in the configuration space $(a, \varepsilon, \mathcal{G})$, we have

$$\oint dI_{\text{PT}} = 0. \quad (13)$$

Therefore the net mechanical work extracted over a complete cycle is exactly zero. This is not an inequality (ordinary first principle) but an *algebraic equality* (anti-double-counting).

8.3 Practical consequences

This property immediately disqualifies several classes of extraction proposals:

1. *Static Casimir cavities with electron ratchets*: the cavity being fixed in $(a, \varepsilon, \mathcal{G})$, the system reaches detailed equilibrium. No net DC current can persist without external injection. The purely geometric ratchet reproduces the Smoluchowski-Feynman error [12]: the apparent thermodynamic asymmetry between wall and antenna is compensated by the detailed balance of tunneling rates.

2. *Material-modulation cycles* (e.g., Pinto-type schemes [13]): the work required to modify $\varepsilon(\omega)$ always exceeds the Casimir gain, because the modulation is itself a path in (ε) that contributes to integral (13).
3. *Stationary vacuum-to-current conversion*: the stationary state imposes $\partial_t I_{\text{PT}} = 0$, hence no flux of persistent information traverses the system. The PT construction does not allow stationarity with extraction.

The only *conversion* (not creation) mechanism compatible with PT is the dynamic Casimir effect (DCE) observed by Wilson *et al.* [11]: the rapid mechanical modulation of a plate converts virtual modes into real photons. The energy budget is strict: $\hbar\omega_{\text{photon}} \leq E_{\text{mechanical injected}}$ at each cycle.

8.4 The architectural false hope

A final theoretical scenario deserves mention. If one could locally modify μ^* (the fixed point of the sieve, normally $\mu^* = 15$), one would change the Casimir pressure and *a priori* produce $\oint dI_{\text{PT}} \neq 0$. But $\mu^* = 15$ is fixed by arithmetic self-consistency (Theorem T5 of [14]): it is not a dynamical parameter but the unique stable fixed point of the sieve flow. Modifying it would amount to modifying arithmetic itself, which is impossible by construction in PT.

8.5 Synthesis

The PT reading does not free the Casimir effect from thermodynamic constraints: it makes them more rigorous by showing that they are arithmetic in nature, not merely physical. Casimir remains a *probe*, never a *source*.

But it is an exceptionally sensitive probe in the informational sense: the 131 ppm of echo signature constitute the most direct observable trace that the PT classification of primes possesses a structural reality beyond the Standard Model. The experimental gain is informational, not energetic.

9 Discussion

Conjecture 1 reframes the Casimir effect within the PT ontology. Three structural observations follow.

9.1 Casimir as probe of the sieve, not of the vacuum

The Casimir effect is not a manifestation of the quantum vacuum but a measurement of the spectral deficit of the constrained CRT sieve. No infinite zero-point energy sum is required, no regularization needed: the Euler product in the PT decomposition is finite at every step. The standard analytic continuation $\zeta_R(s)$ becomes a convenient calculational tool rather than an ontological mystery.

9.2 Ghost primes become observable

The echo tail $\prod_{p \geq 11} (1 - p^{-(d+1)})^{-1}$ is, by Proposition 1, the entire content of the deviation between the PT-native truncation and the standard coefficient. Any sub-permille Casimir measurement therefore tests directly the PT classification of primes into actives, parity, and echoes. This connects the Casimir effect to the cosmological dark-sector identification developed in [14], where echo primes account for the dark matter fraction $F_{\text{echo}} \approx 95\%$ at large N .

The identification is consistent: these are not *two* distinct structures (dark matter *and* Casimir echoes) but *one*, observed at two different scales (cosmological vs. laboratory).

9.3 The prefactor reduces to dimensional counting

The factor $\mathcal{G}_3 = 3/(8\pi^2)$ contains no free parameter once the PT classification is established ($N_{\text{spatial}} = 3$, $N_{\text{transverse}} = 2$). The π^2 comes from the $\mathbb{Z}/p\mathbb{Z} \rightarrow S^1$ limit of the discrete cycle, intrinsic to the PT construction. No fitting is left.

9.4 Comparison with α_{EM}

A similar PT decomposition has been demonstrated for the fine-structure constant [14]: $\alpha_{\text{EM}}^{-1} = 137.0359990834$ from a PT cascade product over actives $\{3, 5, 7\}$, with echo VP corrections from $\{11, 13\}$. The structure

$$\alpha_{\text{EM}^{\text{dressed}}}^{-1} = \alpha_{\nu}^{-1} + \text{echo corrections} \quad (14)$$

mirrors the Casimir structure

$$P_{\text{exact}} = P_{\text{PT}} \cdot \zeta_{\text{echo}}(d+1). \quad (15)$$

Both are observables built from a PT-active core plus echo-prime corrections. The relative magnitude of the echo correction is much smaller in Casimir ($\sim 10^{-4}$ in 3D) than in α_{EM} ($\sim 10^{-3}$) because Casimir lives in the rapidly converging $\zeta(4)$ rather than the slowly converging cascade product of α_{EM} .

9.5 Position in the PT programme

This article completes a trajectory begun by the arithmetic derivation of α_{EM} and extended by the PT_RH [15] chantier on the Riemann zeta function and PT_HL [16] on the Hardy-Littlewood conjectures. Other PT predictions testable at the laboratory scale exist (notably a_{μ} and a_e via echo-VP corrections to the running of α_{EM} [14], both compatible with current measurements). What makes Casimir distinctive is that it is the most *direct and accessible* PT test: the echo signature is isolated arithmetically (not buried in QED running), the prefactor reduces to dimensional counting (zero analytic input), and the prediction sits at a precision level reachable by purpose-built differential measurements in current university laboratories. Other prospective predictions (cosmology: testable by Euclid 2028, DESI 2030, JUNO 2027; negative: no axion, no SUSY < 100 TeV, no proton decay) are either delayed or unfalsifiable in the short term.

Proposition 1 jointly with Theorem 1 positions Casimir as the most accessible experimental crucial test of the PT prime classification.

10 Conclusion

We have established that the standard ideal-plate Casimir pressure coefficient $\pi^2/(240a^4)$ admits a PT-native derivation with zero adjusted parameter. The construction has three components:

1. The geometric prefactor $\mathcal{G}_3 = N_{\text{spatial}}/(2(2\pi)^{N_{\text{transverse}}})$, identified entirely from the PT count of spatial dimensions.
2. The Euler product $\zeta(4) = \prod_p (1 - p^{-4})^{-1}$, decomposed along the PT classification of primes into parity, actives, and echoes.
3. The C3 truncation to $\{2, 3, 5, 7\}$ predicts the standard coefficient to 131 ppm, the residue being rigorously the echo Euler tail $\prod_{p \geq 11} (1 - p^{-4})^{-1}$.

The construction is universal across the spatial dimensions tested ($d \in \{1, \dots, 5\}$), the cleanness of the PT prediction growing super-exponentially with d . The position of 3D Casimir at the *experimental sweet spot* (131 ppm) makes it the most accessible probe of the PT classification.

Jointly, the fundamental identity $\oint dI_{PT} = 0$ excludes any net energy extraction device with a static cavity: the PT reading reinforces thermodynamic constraints by reinterpreting them as exact arithmetic constraints.

A differential Casimir measurement at sub-100 ppm precision would constitute, depending on its result, either the first direct detection of echo primes as a real physical structure, or a falsification of the PT classification. In both cases, the result would be decisive.

Acknowledgments

The author thanks the open-source PT toolkit (PT_CHEMISTRY, PT_PHYSICS, https://github.com/Igrekess/PT_CHEMISTRY) for the computational infrastructure. The numerical results of this paper are reproducible via the scripts `run_p2_coefficient_routes.py` and `run_p2_dimensional_scan.py` in the `PT_EXPERIENCES/PT_CASIMIR/scripts/` directory.

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